

SAE Features Are Causally Active but Predictively Redundant for Difficulty: Evidence Across Language and Time-Series Foundation Models

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github.com/nabindev3/fm-difficulty-probe

Abstract

Foundation models ship without a native abstention signal, yet some inputs are far harder than others. We ask whether a model’s internal TopK sparse-autoencoder (SAE) features encode a *self-difficulty* signal beyond a cheap, no-forward-pass baseline, and we ask it of two unrelated modalities through a single shared pipeline: an autoregressive language model (Pythia-410M on HellaSwag and SQuAD) and an encoder-based time-series foundation model (Chronos-T5-small on ETTh1). Across both modalities, SAE features add *no* incremental predictive power for difficulty over the strongest cheap baseline ($\Delta\text{AUROC}(\text{SAE}=\text{raw}) \leq 0$; label-permutation $p \leq 0.03$, and $p < 10^{-4}$ on SQuAD), and the result is invariant to the SAE expansion factor ($4\times$ vs. $8\times$). The deployable consequence also replicates: a Platt-recalibrated selective predictor built on the cheap baseline captures 30–41% of the oracle’s risk–coverage improvement and yields a Pareto-dominating cost–quality cascade in both modalities. The causal picture, however, *dissociates*: under reconstruction-patching ablation the language model’s top features are causally active (5/5 significant under all-position intervention, 0–2/5 under single-position—coverage, not fidelity), whereas the time-series model’s features are causally silent under either intervention coverage (0/5). The cross-modal dissociation—predictive null in both, causal positive in the LM only—localizes the causal difficulty signal as a property of the autoregressive language model rather than a universal property of sparse features. We release a modality-agnostic, regression-gated, one-command-reproducible implementation.

1 Introduction

Foundation models (FMs) are deployed as black boxes: a language model (LM) generates an answer, a time-series foundation model (TSFM) emits a forecast, and in both cases every input consumes the same compute and is granted the same trust, even though some inputs are several times harder than others. Knowing *which* inputs a model will struggle on—before paying for an expensive model or committing to a forecast—is the core of routing, abstention, and cost–quality cascades [5, 9].

A natural hypothesis, given the rise of mechanistic interpretability, is that an FM’s internal representations already encode this self-difficulty signal, and that sparse autoencoders (SAEs)—which decompose activations into interpretable, monosemantic features [4, 6]—expose it in a usable form. The open, and to our knowledge unclaimed, question is whether SAE features add difficulty signal *beyond what a cheap baseline already captures*: lexical statistics for text, classical descriptive statistics for a time series. A single negative answer on one model invites the reflex “you just probed it wrong.” We therefore ask the question of two modalities *through the same pipeline, with*

the same metrics, so that what replicates is a property of SAEs and what does not is localized to one architecture.

Contributions. (i) A modality-agnostic pipeline—one tested core, two thin adapters—that runs the identical probe ladder, causal ablation, and selective-prediction analysis on an LM and a TSFM, gated on reproducing both prior studies’ headline numbers before any new experiment is trusted (Section 2). (ii) A *predictive null* that replicates across modalities and is robust to SAE width and to a label-permutation test (Sections 3–5). (iii) A *causal dissociation*: the same reconstruction-patching ablation finds the LM’s features causally active but the TSFM’s features silent, and on the LM side localizes per-feature detectability to intervention *coverage*, not SAE fidelity (Section 5). (iv) A deployable artifact—a Platt-recalibrated selective predictor on the cheap baseline—that replicates in both modalities (Section 6).

2 Method: one pipeline, two modalities

We describe the pipeline once; the two backends differ only in an adapter that implements a common **Modality** interface (cheap features, raw activations, SAE codes, labels, splits, CV scheme, causal metric, cascade models). Nothing in the shared core imports a specific model.

Backbones and data. The LM modality uses Pythia-410M [3] (residual stream at Layer 12 “mid” and Layer 18 “late”), evaluated on HellaSwag [12] (binary multiple-choice correctness; $N=5000$, 3500/1500 train/test) and SQuAD [11] (continuous gold-answer perplexity difficulty). The TSFM modality uses Chronos-T5-small [2] (encoder block 3 “mid” and block 5 “late”) on ETTh1 [13] (701 forecast windows, 483/167 train/test, context 512, horizon 96). Cascades use Pythia-2.8B and Chronos-T5-base as the expensive model.

Sparse autoencoder. A TopK SAE with $k=32$ and auxiliary dead-feature revival is trained on the *train split only*. The LM SAE uses $4\times$ expansion ($d_{\text{hidden}}=4096$), the TSFM SAE $8\times$ ($d_{\text{hidden}}=4096$); we test invariance to this choice in Section 5.

The probe ladder. The crux of an apples-to-apples cross-modal claim is that both modalities report the *same* comparison. We fit an L_1 -regularized logistic probe on five feature sets and rank them: **P1** cheap baseline (lexical stats | classical TS stats); **P2** cheap + raw activations; **P3** cheap + SAE codes; and diagnostic isolations **P4** raw-only, **P5** SAE-only. The headline quantity is $\Delta(\text{SAE} - \text{raw}) = \text{AUROC}(\text{P3}) - \text{AUROC}(\text{P2})$: does the SAE decomposition add anything *on top of* the raw activations it was trained on? Regularization C is chosen by inner cross-validation that respects each modality’s structure (stratified-by-category for the LM, `TimeSeriesSplit` for the TSFM). All deltas carry paired-bootstrap 95% CIs ($B=2000$), and the headline delta additionally carries a label-permutation p -value ($B=10,000$).

Causal ablation. We rank the top-5 difficulty-predictive latents and, for each, measure the change in a continuous metric (negative log-probability / cross-entropy for the LM in nats; CRPS [7] for the TSFM) when that latent is zeroed in a forward-hook SAE reconstruction of the chosen layer, relative to the reconstruction baseline. Crucially, the intervention *coverage* is a first-class knob: *single-position* (one boundary token / last context step) versus *all-position* (every prompt token / every context step), with the metric held fixed so coverage is not confounded with the binary/continuous choice.

Selective prediction, calibration, cascade. Risk-coverage curves report the area under the curve (AURC) and the fraction of the oracle’s improvement captured, on each modality’s natural error scale (binary correctness for the LM, continuous CRPS for the TSFM). Probe probabilities are recalibrated with Platt [10] and isotonic regression fit on 5-fold out-of-fold train predictions. A threshold sweep routes between the cheap and expensive model, tracing a Pareto frontier against random and oracle routing.

Leakage controls and the regression gate. Both adapters enforce: train-only SAE fitting, a temporal purge gap $\geq \text{context} + \text{horizon}$ (TSFM) and pretraining-contamination / lexical-dedup filters (LM), a prompt-only perplexity feature that never sees the gold target, and CV that respects category/temporal structure. We adopt a strict *regression gate*: no new experiment is trusted until both prior studies’ headline numbers reproduce through the shared code—LM SQuAD/L18 raw-only AUROC = 0.716 and TSFM $\Delta(\text{SAE} - \text{cheap}) = -0.228$, both reproduced exactly. The probe refuses to run on a missing or randomly-initialized SAE checkpoint and on single-class labels.

3 Results A — Language model (Pythia)

Predictive null. On SQuAD, raw residual activations are genuinely useful—P2 reaches AUROC 0.671 at Layer 12 and 0.716 at Layer 18, versus 0.590 for lexical stats—but the SAE adds nothing on top: $\Delta(\text{SAE} - \text{raw}) = -0.079$ (95% CI $[-0.118, -0.041]$), with a label-permutation $p < 10^{-4}$. HellaSwag is a near-chance regime for every probe family (P1 0.509, P3 0.500), so no method adds signal there. In neither case does the SAE decomposition beat the raw activations.

Causal positive, and coverage-not-fidelity. The same features the probe ranks are causally active: under all-position patching, 5/5 of the top SQuAD features (and 5/5 HellaSwag) show ablation effects whose CIs exclude zero. Under single-position patching the count collapses to 0–2/5—the effect is real but *below the resolution of a single-token intervention*. Per-feature causal attribution requires multi-position coverage, not a higher-fidelity SAE.

4 Results B — Time-series model (Chronos-T5)

Predictive null, more starkly. On ETTh1 the cheap classical statistics are the *strongest* probe (P1 0.654); raw encoder activations do worse (P2 0.584) and the SAE worse still (P3 0.426). Thus $\Delta(\text{SAE} - \text{cheap}) = -0.228$ (CI $[-0.366, -0.092]$, the legacy headline) and $\Delta(\text{SAE} - \text{raw}) = -0.158$ (CI $[-0.293, -0.025]$, $p = 0.011$). The null holds at both encoder layers.

Causal silence. Under the same reconstruction-patching protocol at full sample budget (50 samples/window, all 167 test windows), *no* feature is significant under either all-position or single-position intervention (0/5 both), reproducing the prior Chronos null. The TSFM’s features are predictively redundant *and* causally quiet at this scale.

5 Cross-modal synthesis

Table 1 places the two modalities side by side. Two of the three legs of the going-in thesis replicate; one dissociates—and the dissociation is the contribution.

Table 1: Cross-modal synthesis. Predictive null and deployable selective predictor replicate in both modalities; the causal contribution is specific to the language model. Δ AUROC CIs are paired-bootstrap ($B=2000$); p is a one-sided label-permutation test ($B=10,000$).

Axis	LM (Pythia)		TSFM (Chronos)
	HellaSwag	SQuAD	ETTh1
P1 cheap AUROC	0.509	0.590	0.654
P2 cheap+raw AUROC	0.472	0.671	0.584
P3 cheap+SAE AUROC	0.500	0.592	0.426
$\Delta(\text{SAE} - \text{raw})$	+0.028	−0.079	−0.158
95% CI	[−0.001, .058]	[−0.118, −0.041]	[−0.293, −0.025]
label-perm p (1-sided)	0.033	$< 10^{-4}$	0.011
causal: all-position	5/5	5/5	0/5
causal: single-position	0/5	2/5	0/5
selective: % oracle AURC	2.0%	41.3%	30.5%
cascade: Pareto-dom. points	1	31	5

The one-sentence claim. Across an autoregressive LM and an encoder TSFM, TopK-SAE features are predictively redundant for difficulty beyond the strongest cheap baseline ($\Delta(\text{SAE} - \text{raw}) \leq 0$, $p \leq 0.03$) and a cheap-baseline selective predictor is deployable in both (30–41% of oracle AURC), *but* the causal contribution is specific to the LM (5/5 features significant under all-position patching vs. 0/5 on Chronos under either coverage).

Robustness to SAE width. Re-training each modality at the other’s expansion (LM $4\times \rightarrow 8\times$, TSFM $8\times \rightarrow 4\times$) and re-probing leaves the null intact: $\Delta(\text{SAE} - \text{raw})$ stays significantly negative on SQuAD ($-0.079 \rightarrow -0.058$) and ETTh1 ($-0.158 \rightarrow -0.100$). The *larger* LM SAE still does not beat raw, directly answering “maybe a bigger SAE would have found signal.”

Why the dissociation matters. If the TSFM cells had echoed the LM cells, the contribution would be a cross-modal replication. Because they diverge, the contribution is sharper: the causal difficulty signal is not a universal property of sparse features but a property of *this* autoregressive LM, while the *predictive* null and the *deployable* positive are universal. Prior work did one half each on a single modality [1, 8]; the dissociation requires both halves on both modalities through one pipeline.

6 Deployable artifact

When the answer to “do internals help?” is “no, but the cheap baseline does,” the cheap baseline is still deployable. A Platt-recalibrated selective predictor on the cheap rung captures 41.3% (LM, raw scale) and 30.5% (TSFM) of the oracle’s AURC improvement; recalibration on 5-fold out-of-fold predictions cuts expected calibration error substantially under Platt scaling, which preserves the ranking (e.g. SQuAD P1 ECE $0.366 \rightarrow 0.123$; ETTh1 P1 $0.284 \rightarrow 0.156$; isotonic reaches 0.119 on ETTh1 but is non-monotone). Routing on the calibrated score yields Pareto-dominating cost–quality points in both modalities (31 on SQuAD, 5 on ETTh1), beating always-cheap, always-expensive, and random routing.

7 Limitations

We probe a single SAE family (TopK) at two expansions; a different dictionary-learning objective could in principle expose signal TopK does not, though the width sweep argues against a capacity explanation. We use two model scales per modality and one dataset per LM task and one TSFM dataset; the near-chance HellaSwag regime limits what can be concluded there. The causal n on the TSFM is bounded by the 167-window test set; the 0/5 result is a null, and nulls are weaker than positives, though it reproduces at full sample budget. Finally, “difficulty” is operationalized per modality (binary correctness / top-quartile CRPS); other operationalizations may behave differently.

8 Conclusion

The same sparse features that a probe ranks as “difficulty-predictive” add no predictive power over raw activations in either a language or a time-series foundation model, yet are causally active in the language model alone. The deployable lesson is modality-invariant—recalibrate the cheap baseline—while the scientific lesson is sharp: a causal self-difficulty signal in sparse features is, at these scales, a property of the autoregressive language model and not of foundation models in general.

Reproducibility. All numbers and figures regenerate from one command (`reproduce.sh`); the shared core is unit-tested on synthetic arrays and contains no model-specific code. Code: <https://github.com/nabindev3/fm-difficulty-probe>.

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A Per-modality detail

The LM adapter (Pythia) and TSFM adapter (Chronos) each own their feature extractors, hooks, CV scheme, and causal metric; see the per-modality repositories <https://github.com/nabinddev3/llm-sae-difficulty> and <https://github.com/nabinddev3/tsfm-sae-difficulty> for the full leakage-control ablations, the 2×2 coverage/fidelity disentanglement on the LM, and the selective-forecasting calibration curves on the TSFM.